

Programmer Assessment – Operations System + AI Challenge

Company Overview

Company: Sejuk Sejuk Service Sdn Bhd (*fictional name*)

Sejuk Sejuk Service provides **air-conditioner installation, servicing and repair** for homes and businesses.

Operational context:

- 5 branches nationwide
- 40+ technician teams operating in the field
- Admin staff mainly work on **desktop systems**
- Technicians primarily use **mobile devices in the field**

Business Goal

Digitise the full service workflow:

Order → Assignment → Service Completion → Manager / Accounts Review

This assessment simulates a **simplified internal operations system** used by the company.

What's This Assessment About?

We want to understand how you:

- Build **real-world business workflows**
- Think in **systems and data models**, not just pages
- Design **clean and practical interfaces**
- Integrate **AI capabilities into business tools**
- Explain your technical decisions clearly

You may complete **one or more modules**.

We care more about **how you think and design the system** than the number of modules completed.

Suggested Tech Stack

Candidates may use any stack they prefer.

However, our preferred stack is:

Layer	Preferred Tools
Front-end	React
Styling	Tailwind CSS
Backend / Database	Supabase
File Storage	Supabase Storage
Deployment	Vercel
Login	Simple mock login / role switch

Authentication

Real authentication is **not required**.

A simple mock login or role selector is sufficient.

Example roles:

- Admin
- Technician
- Manager

You may implement a simple role switch to simulate different users.

Bonus if you implement basic authentication.

AI Integration

For AI-related modules, candidates may use:

Layer	Preferred Tools
AI API	OpenAI / Claude / Gemini / equivalent
Backend AI logic	Node.js / Serverless function
Data retrieval	Supabase queries / backend functions

The AI assistant should answer questions **based on system data retrieved through controlled queries**.

It should **not rely on unrestricted access to the entire database**.

AI responses should be based on structured data retrieved from the system.

Simplified System Workflow

Admin creates an order



Technician receives job



Technician completes service



System sends notification



Manager reviews job



Dashboard updates performance metrics

Basic System Rules

Orders follow a simplified workflow state:

New → Assigned → In Progress → Job Done → Reviewed → Closed

Example rules:

- Only **Admin** can assign technicians
- Only the **assigned technician** can mark a job as completed
- Managers may review completed jobs
- Key actions should be **traceable**

You do not need to fully enforce these rules but your system should **consider them in design**.



Module 1 — Admin Portal · Order Submission

Goal: Admin creates a service order and assigns a technician

Section	Details
Form fields	Order No (auto-generated) Customer Name Phone Address Problem Description Service Type (dropdown) Quoted Price Assigned Technician Admin Notes
Key skills	Front-end: responsive form, clean UI Back-end: auto-generated ID + write to database
Bonus	• Show order summary after submission • Send WhatsApp notification to technician



Module 2 — Technician Portal · Service Job

Goal: Technicians view assigned jobs and record completed service.

Important Note

The technician workflow should prioritize speed and simplicity for field usage.

Technicians should be able to complete the job quickly using a mobile device.

Candidates may implement this as:

- a mobile-first web interface
- or a mobile app

Section	Details
Form fields	Order ID (read-only) Work Done Extra Charges Upload ≤ 6 photos / video / PDF Final Amount (auto-calculated) Remarks Technician Name Timestamp Optional Bonus: Technician may record payment received from customer <i>Example fields:</i> Payment Amount Payment Method Receipt Photo
Key skills	Front-end: mobile-friendly UI, simple field workflow Back-end: file upload, writing service completion data
Bonus	• Generate WhatsApp feedback message send to customer • Notify manager/accounts when job completed

Sample Data (JSON)

```
{  
  "orderId": "ORDER1234",  
  "customerName": "Ahmad",  
  "address": "No. 12, Jalan Sejuk, Shah Alam",  
  "service": "Aircond cleaning",  
  "assignedTechnician": "Ali",  
  "status": "Pending"  
}
```

Mock technicians: **Ali, John, Bala, Yusoff**

Module 3 — WhatsApp Notification Trigger

Goal: Send WhatsApp notification when a job is marked **Job Done**.

- **Trigger Condition:** status = *Job Done*
- **Notification Methods:** WhatsApp (preferred) — e.g. a deep-link URL with a pre-filled message

Sample message

Hi {{Customer Name}},
Job {{Order ID}} has been completed by Technician {{Name}} at {{Time}}.
Please check and leave feedback.
Thank you!

Key skill: back-end trigger logic + API / deep-link integration

Bonus Module — KPI Dashboard

Goal: Display technician performance metrics.

Section	Details
Example Metrics	Technician Jobs Completed Total Amount Postpone / Reschedule
UI Suggestions	Card layout Leaderboard Simple charts
Assessment Focus	data aggregation clear visualisation

You may implement metrics using **weekly data** (minimum).

Bonus if additional views are included.



AI Module — Operations Query Window

Goal: Build a simple **AI assistant inside the portal** that allows managers to ask operational questions about service data.

Example questions:

- What jobs did technician Ali complete last week?
 - Which technician completed the most jobs this week?
 - How many jobs were completed today?
-

Expected System Flow

User question



System interprets question/query



Database retrieves relevant data



AI formats the response

Example output:

Technician Ali completed 3 jobs last week:

ORDER1234 – Cleaning

ORDER1237 – Repair

ORDER1241 – Gas refill

Assessment Focus

We want to see:

- how AI integrates with system data
- how queries retrieve relevant records
- how responses are generated clearly
- how limitations are handled

★ Optional Advanced AI Challenges

Candidates may optionally implement one or more of these.

AI Workflow Supervisor

Use AI to flag potential issues in completed jobs.

Example alerts:

Final amount much higher than quoted price

Job done but no photos uploaded

AI Document Understanding

Use AI to extract structured data from uploaded documents.

Example extracted fields:

Customer name

Service type

Service details

Amount

Date

AI Operational Insight

Use AI to analyze operational data.

Example question:

Which technician might be overloaded this week?

Example output:

Technician Bala completed 11 jobs this week, which is significantly higher than the team average.

Optional Self-Assessment (README)

Candidates may include:

- Which module was easiest?
 - Which module was hardest?
 - What would you improve in a real production system?
 - How did you use AI tools while building this project?
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Submission Guide

1. Live demo (Netlify / Vercel / Firebase) **or** ZIP
2. GitHub repo (preferred)
3. README explaining:
 - What you built
 - Tech stack used
 - Architecture decisions
 - Challenges / assumptions
 - How AI was integrated
 - What limitations exist in your implementation?

Also describe:

What types of AI queries are supported

Limitations of your AI implementation

If you only did one module, explain your logic clearly — it helps us see your strengths.

Design & Branding

No logo required.

Focus on:

- clean UI
- simple UX
- practical design

Treat this like a **real-world internal operations tool**.

Estimated Effort

Candidates are not expected to complete everything.

A solid implementation of one or two modules with clear explanation is sufficient.

A well-explained partial implementation is acceptable.
